

EE 3202.304

ECE Fundamentals Lab II

Lab 8 & 9: ADC and DAC

Name: Krish Dhage

Lab Partner: Vishal Manjunath

Dr. Lawrence Overzet

TA: Shideh Tayeb Naimi

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I. Experimental Results

Part 1 – 1 Bit ADC

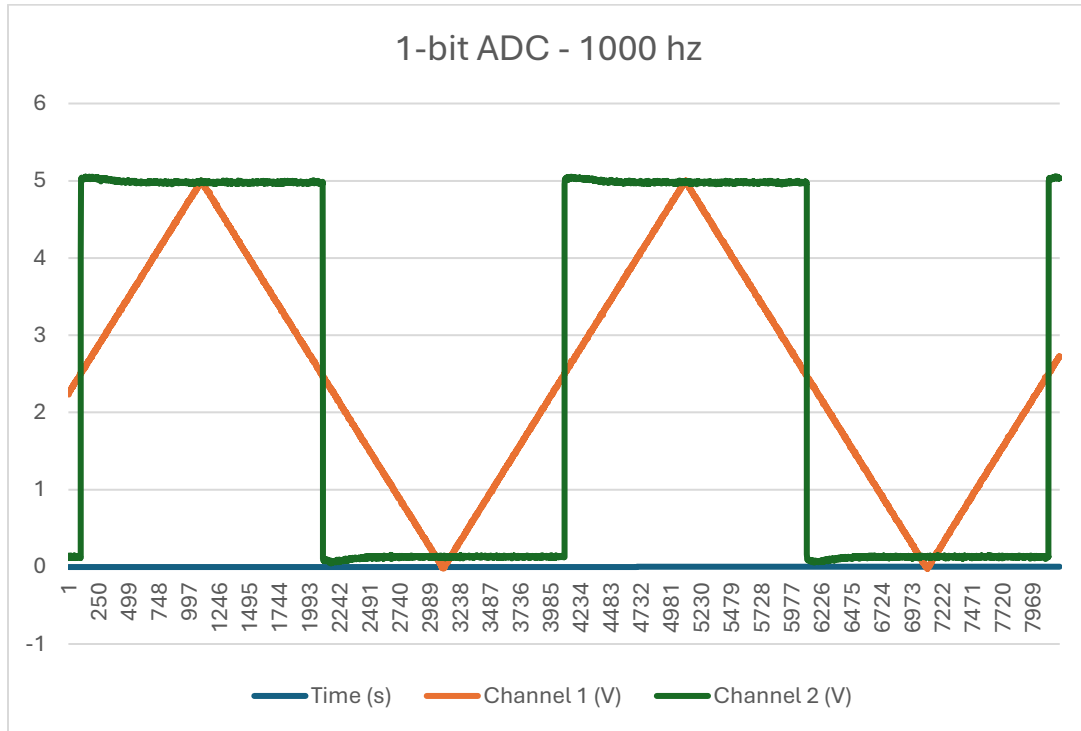


Figure 1.1: 1- bit ADC at 1000 hz

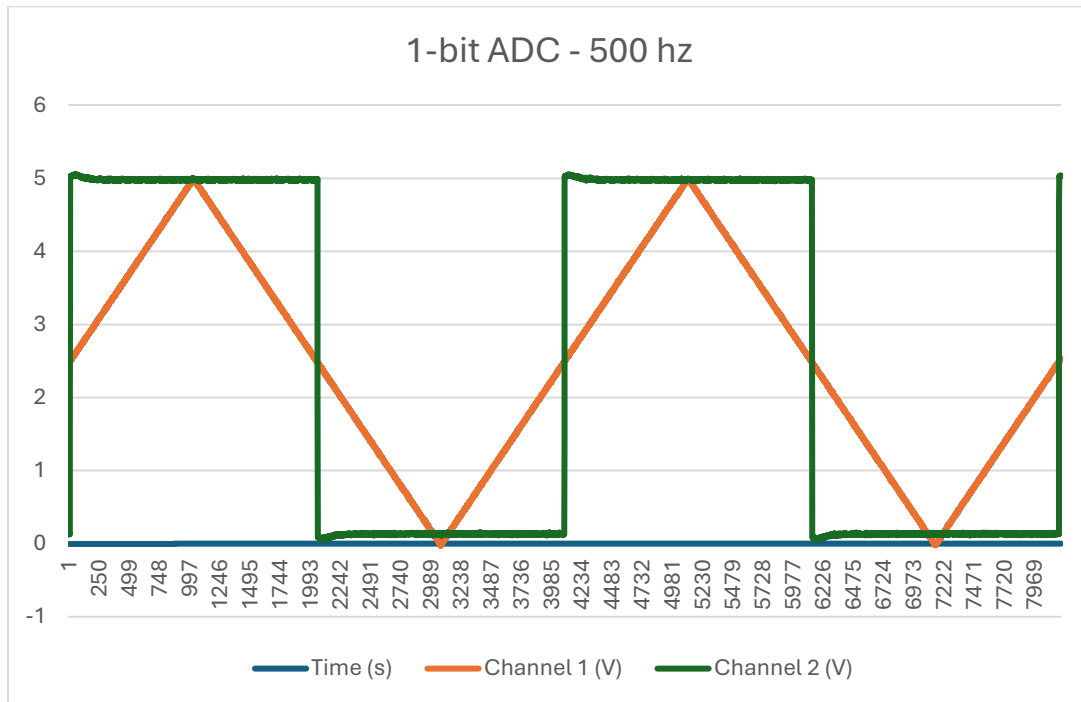


Figure 1.2: 1- bit ADC at 500 hz

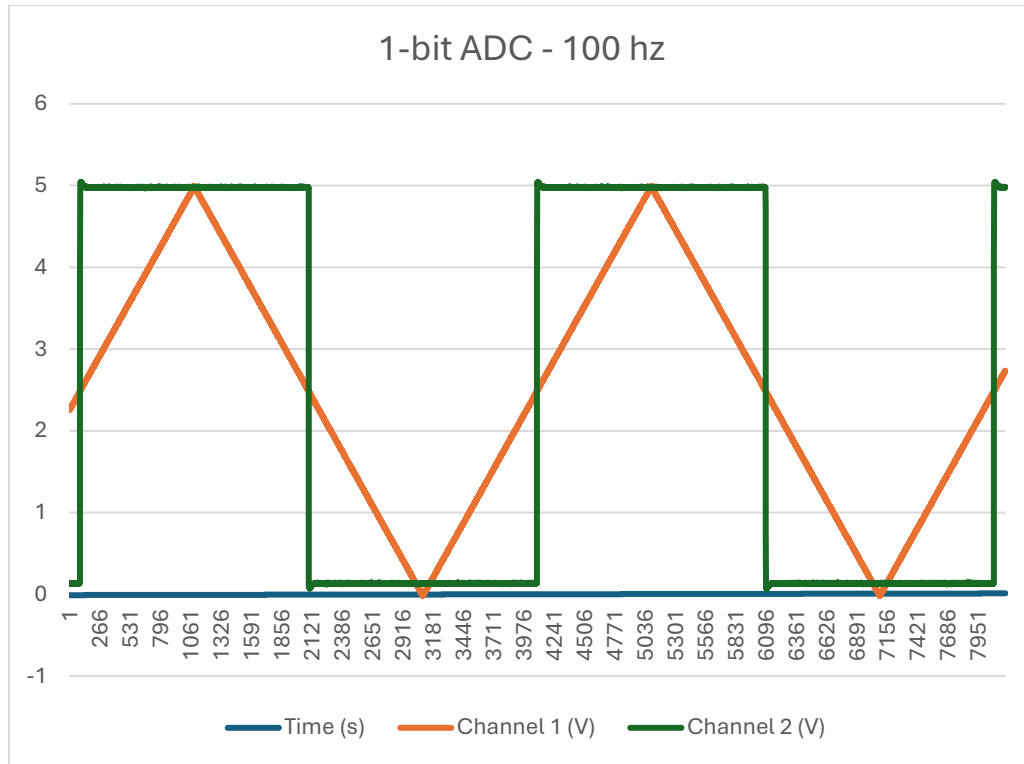


Figure 1.3: 1- bit ADC at 100 Hz

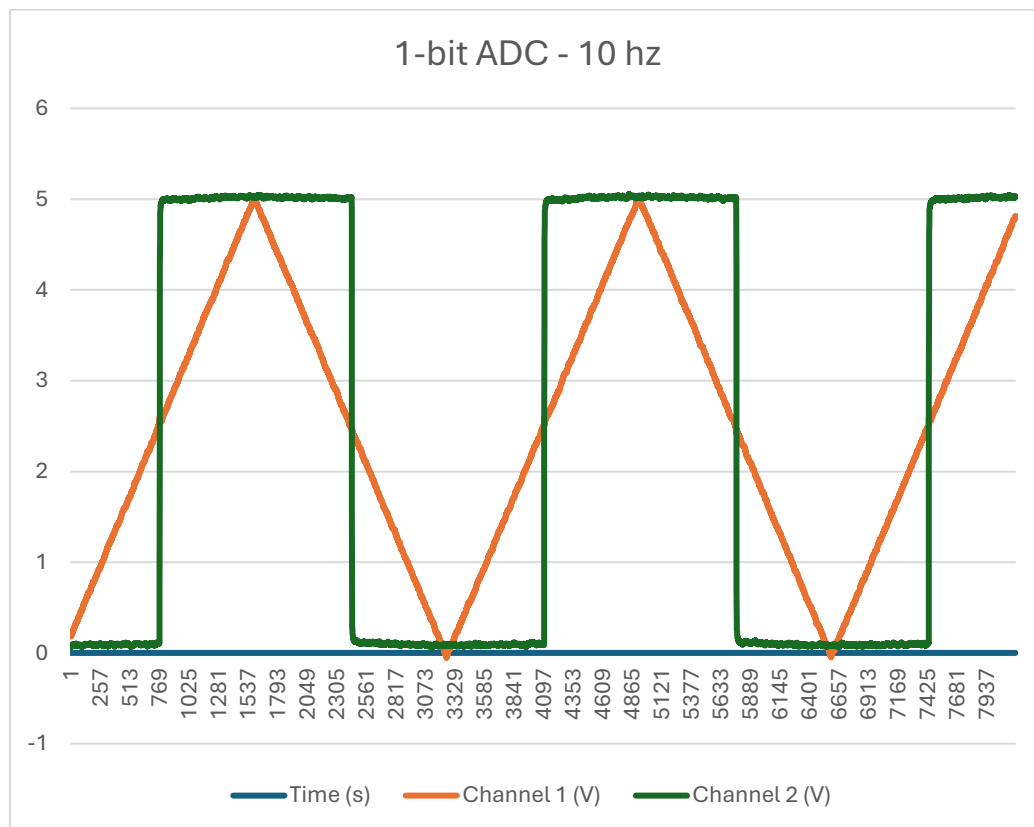


Figure 1.4: 1- bit ADC at 10 Hz

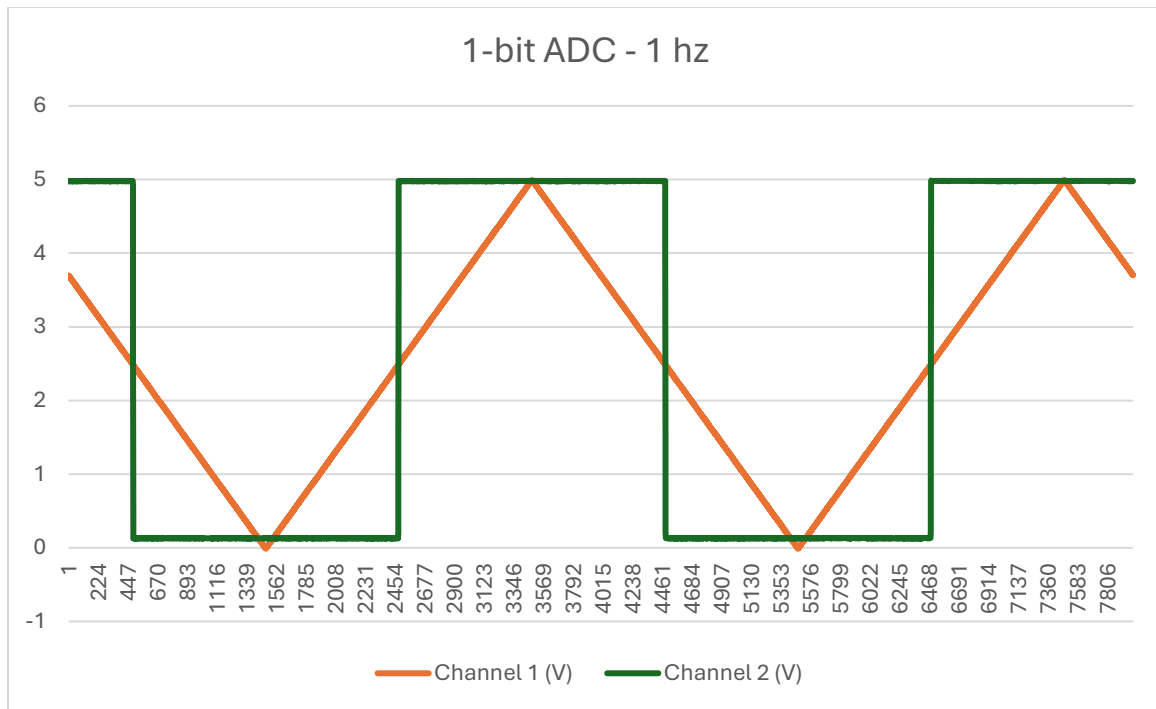


Figure 1.5: 1- bit ADC at 1 hz

Part 2 – 2 Bit ADC

2 Bit ADC : 10kHz

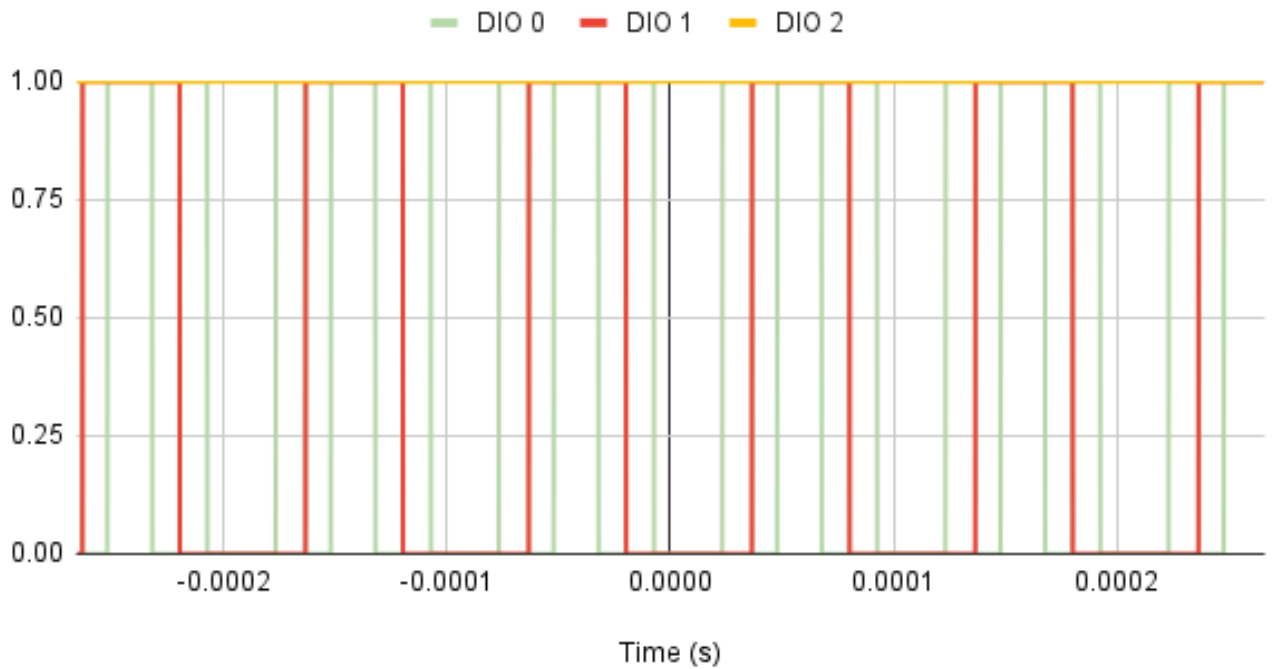


Figure 2.1: 2-bit ADC at 10,000 hz

2 Bit ADC : 1kHz

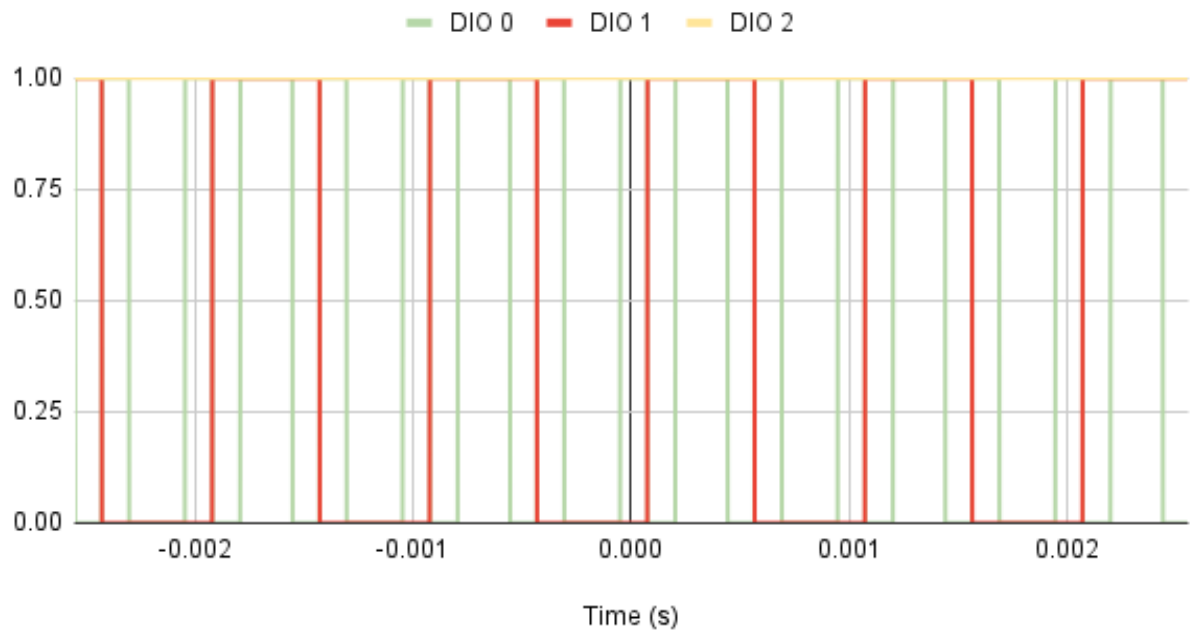


Figure 2.2: 2-bit ADC at 1,000 hz

2 Bit ADC : 500 Hz

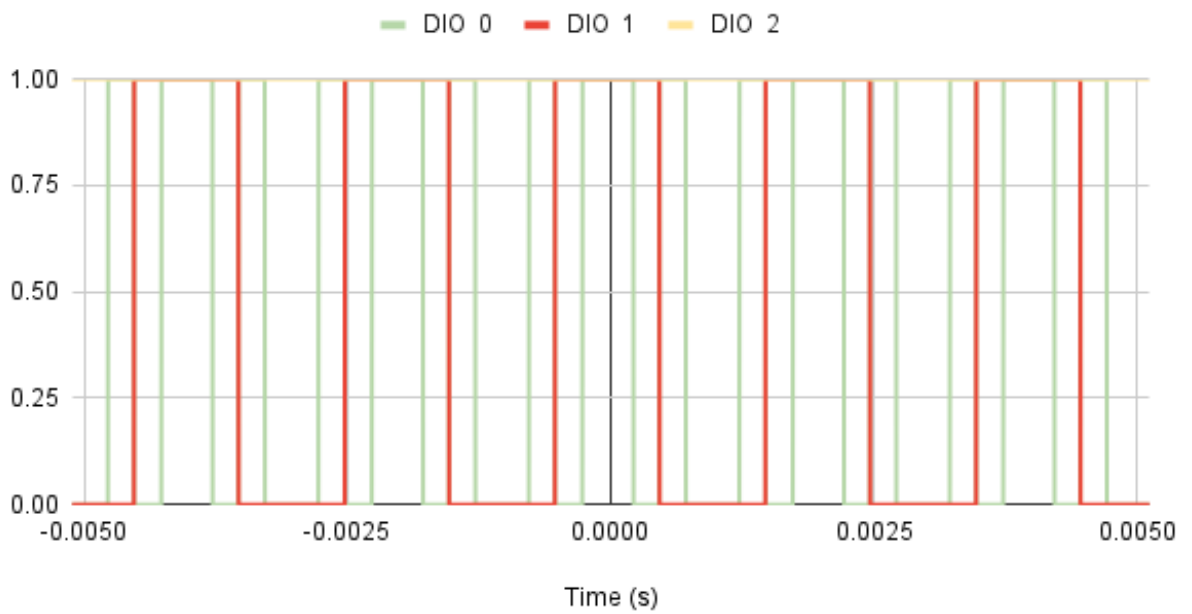


Figure 2.3: 2-bit ADC at 500 hz

2 Bit ADC : 100Hz

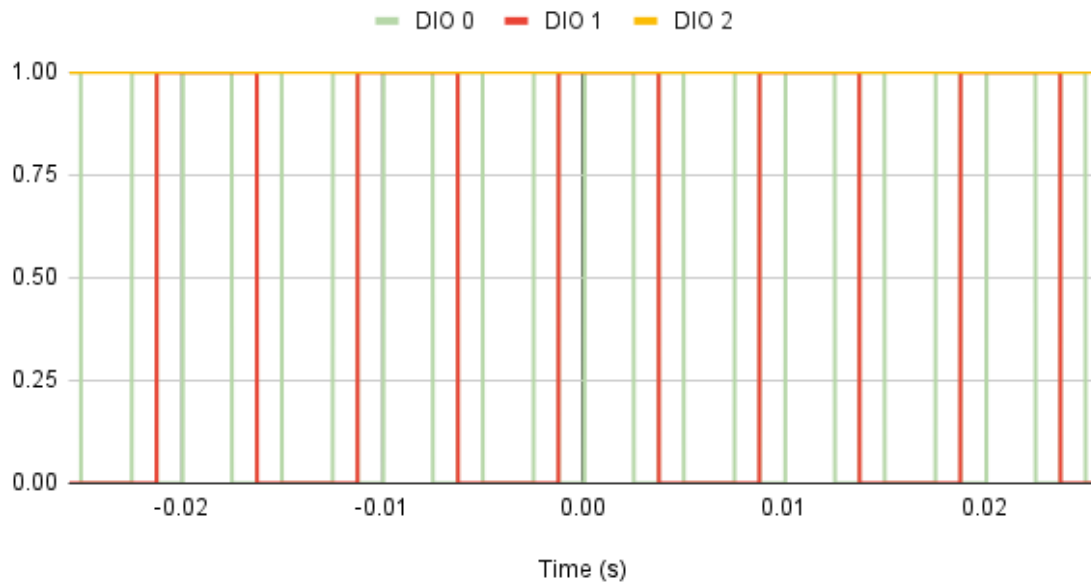


Figure 2.4: 2-bit ADC at 100 hz

2 Bit ADC : 1Hz

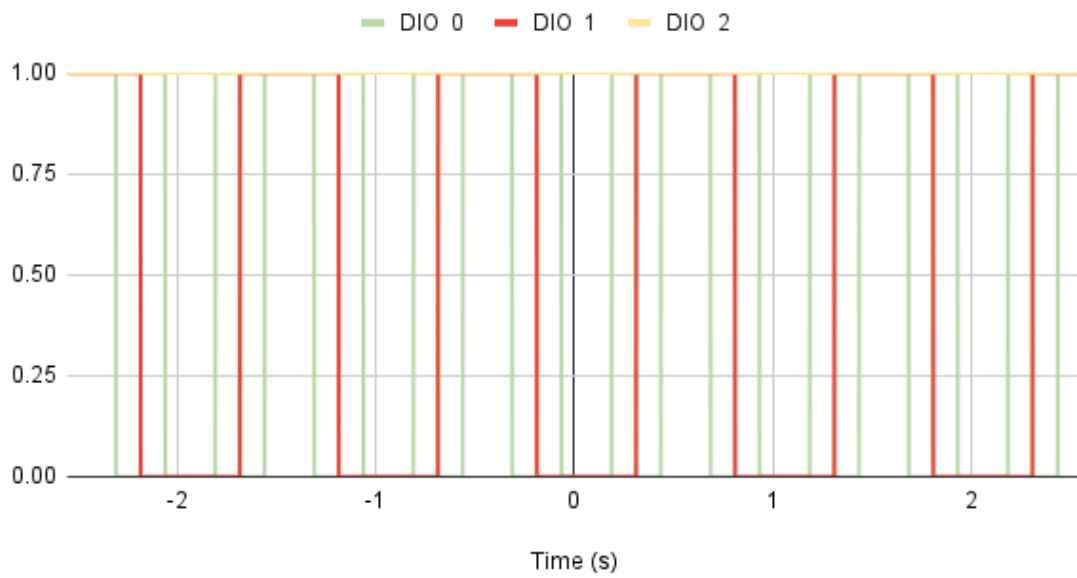


Figure 2.5: 2-bit ADC at 1 hz

Part 3 – RMS Calculations

1-bit ADC:

Step size = 1 V

Integrate the RMS function

$$\text{RMS} = 1/\sqrt{12} = 0.289 \text{ V}$$

2-bit ADC:

Step size = 1/3 V

Integrate the RMS function

$$\text{RMS} = (1/3)/\sqrt{12} = 0.096 \text{ V}$$

Compare 1-bit to the 2-bit:

$$0.289/0.096 = 3.01$$

The 2-bit ADC reduces the error by a factor of 3 compared to the 1-bit RMS. Therefore, the number of bits will significantly affect the error.

Part 4 – 2 Bit DAC Sinusoid Waveform

R-2R Sine : 10kHz

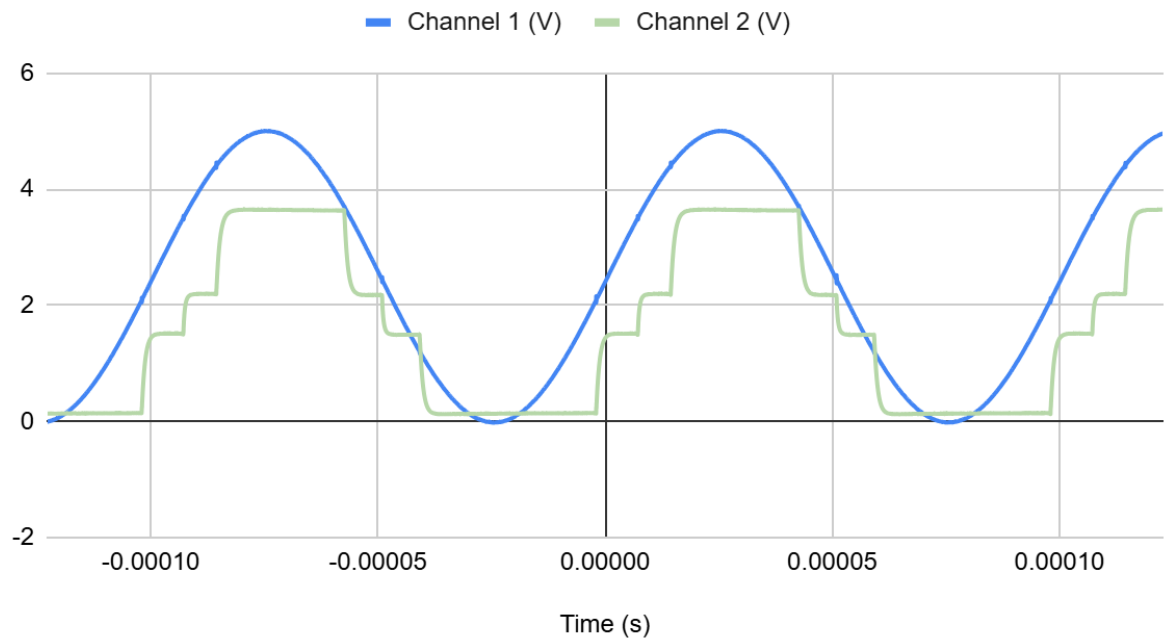


Figure 4.1: 2-bit DAC with Sinusoid Waveform at 10,000 hz

R-2R Sine : 5kHz

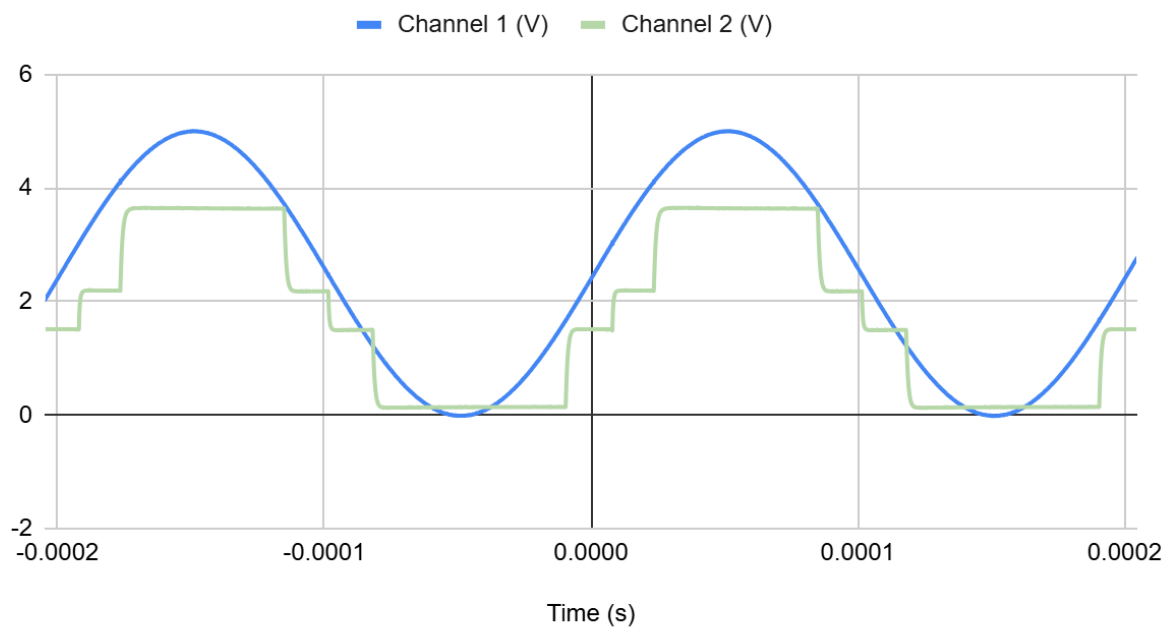


Figure 4.2: 2-bit DAC with Sinusoid Waveform at 5,000 hz

R-2R Sine : 2kHz

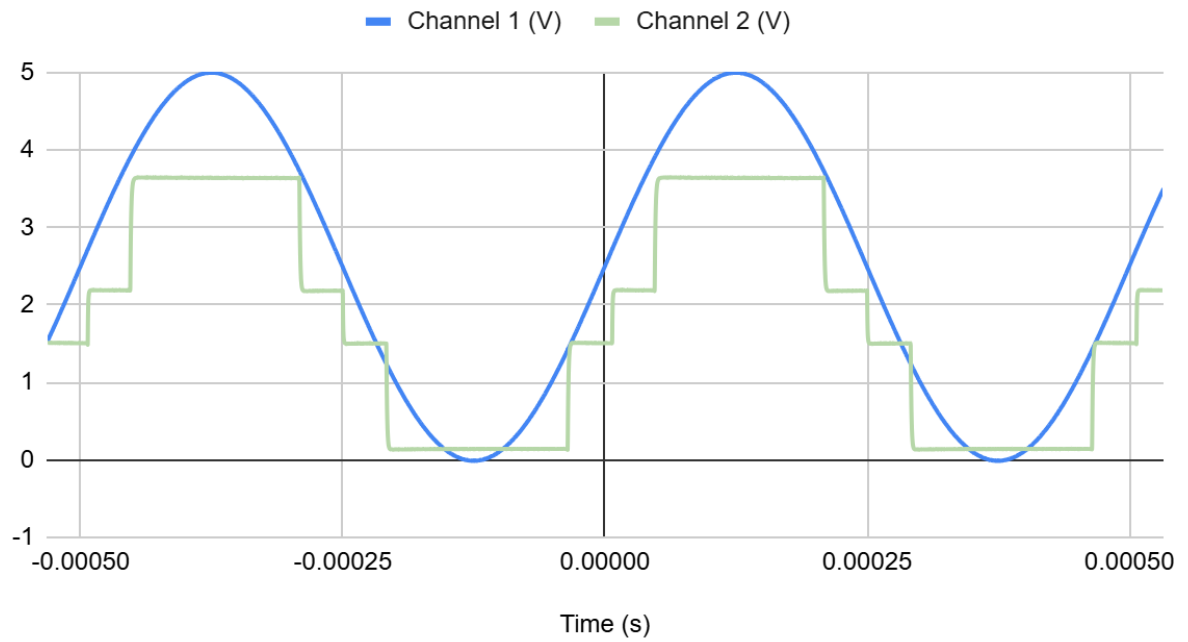


Figure 4.3: 2-bit DAC with Sinusoid Waveform at 2,000 hz

R-2R Sine : 1kHz

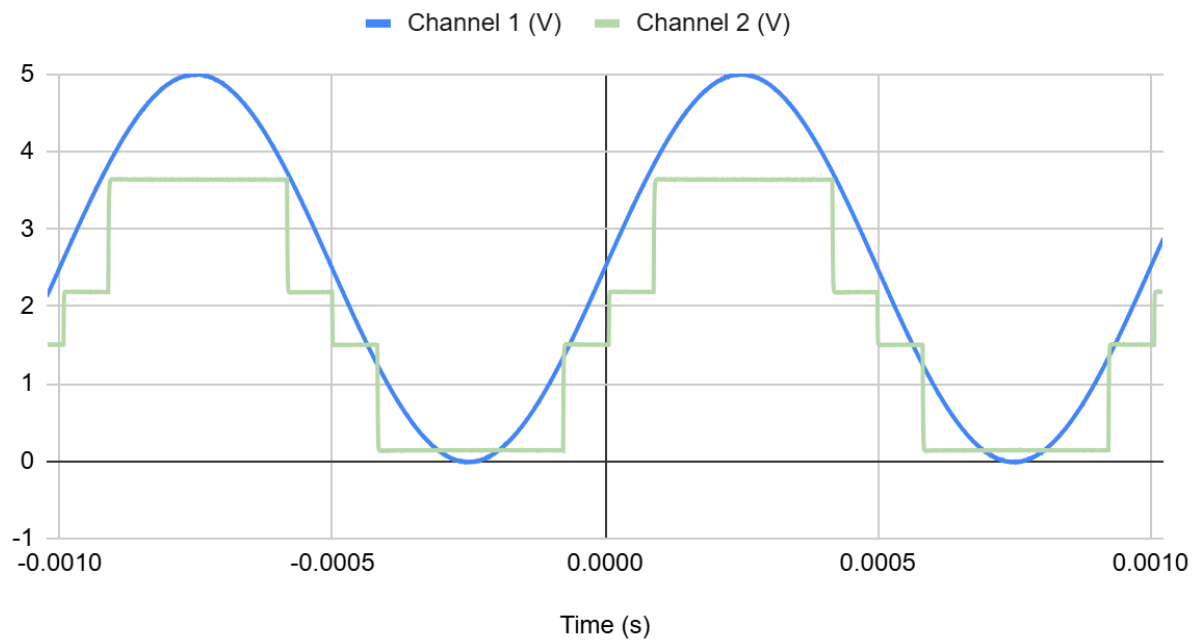


Figure 4.4: 2-bit DAC with Sinusoid Waveform at 1,000 hz

Part 5 – 2 Bit DAC Triangle Waveform

R-2R Triangle Wave : 10kHz

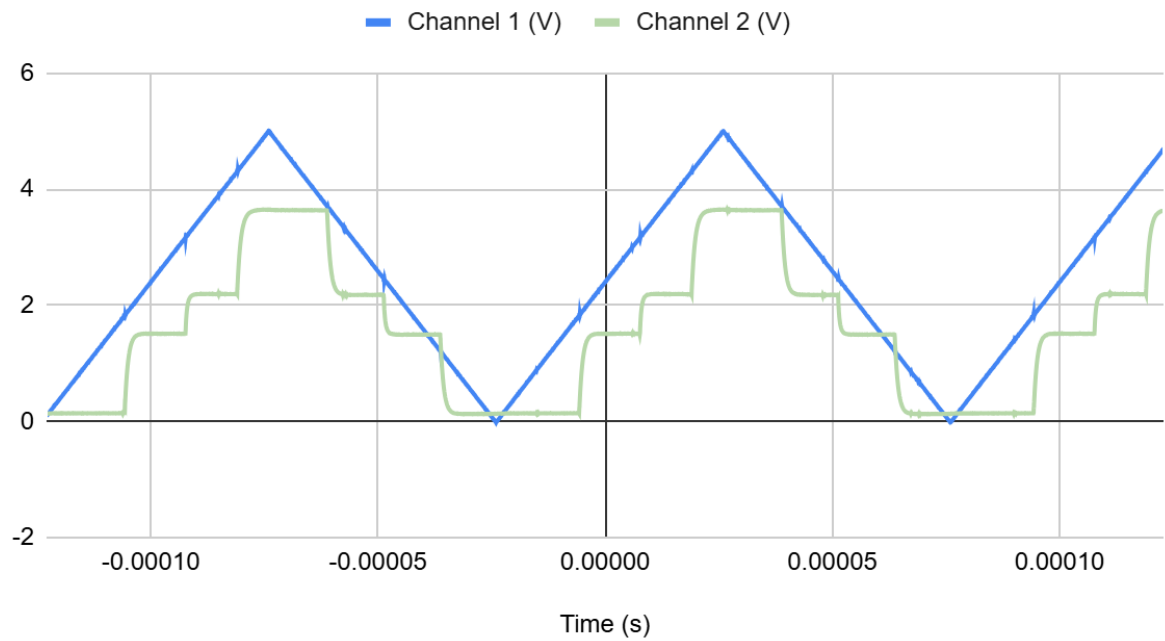


Figure 5.1: 2-bit DAC with Triangle Waveform at 10,000 hz

R-2R Triangle Wave : 5kHz

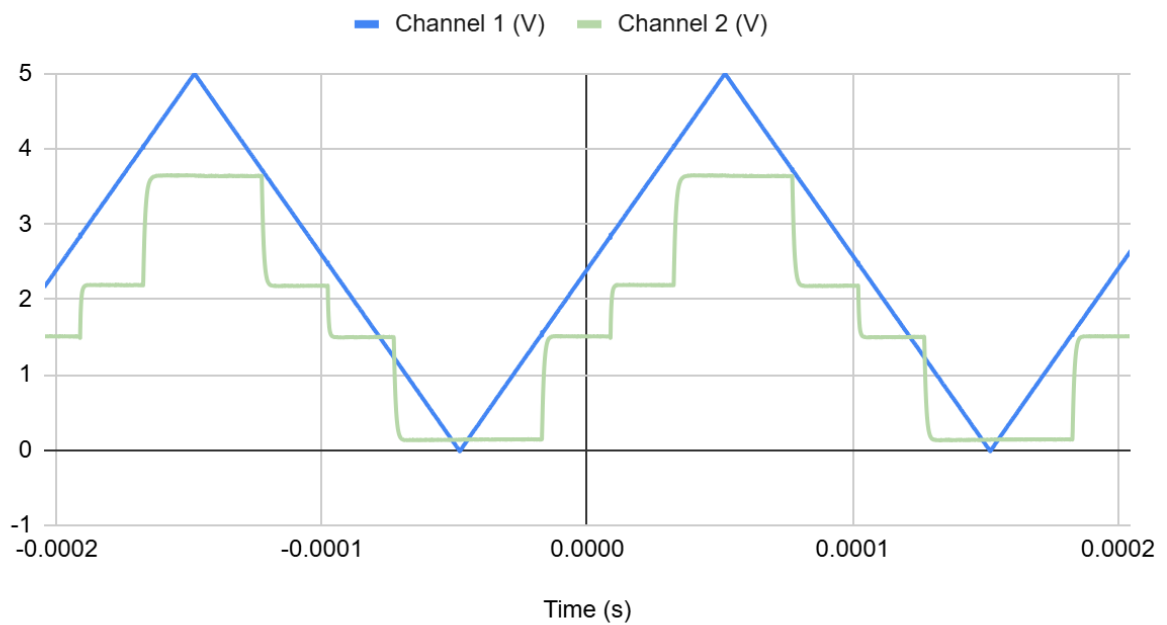


Figure 5.2: 2-bit DAC with Triangle Waveform at 5,000 hz

R-2R Triangle Wave : 2kHz

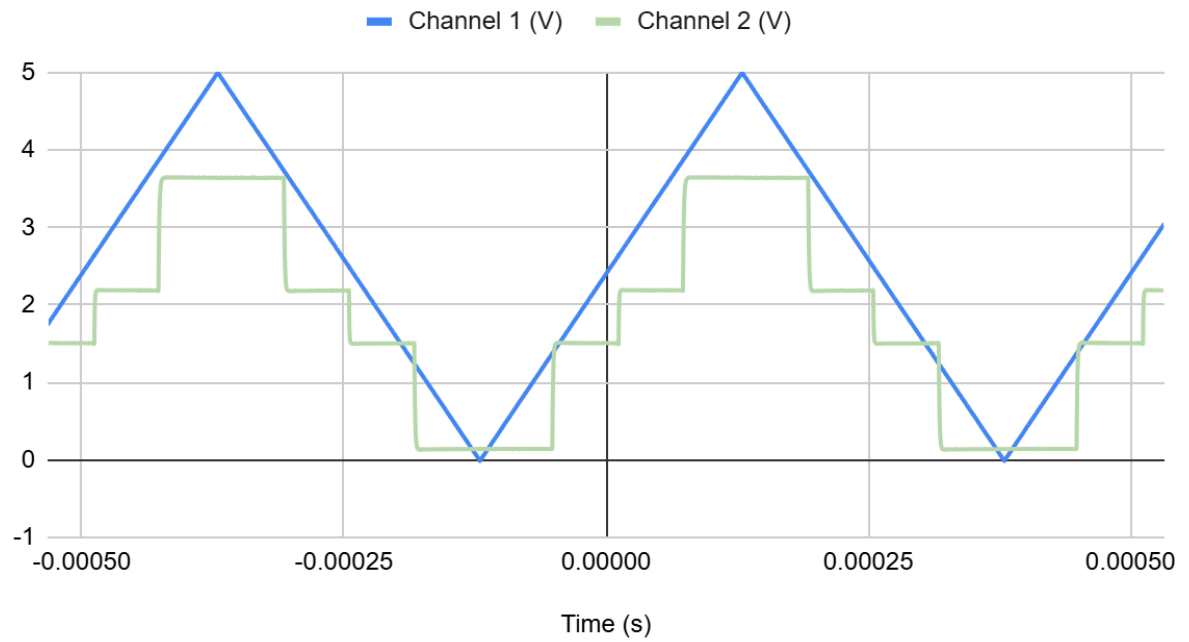


Figure 5.3: 2-bit DAC with Triangle Waveform at 2,000 hz

R-2R Triangle Wave : 1kHz

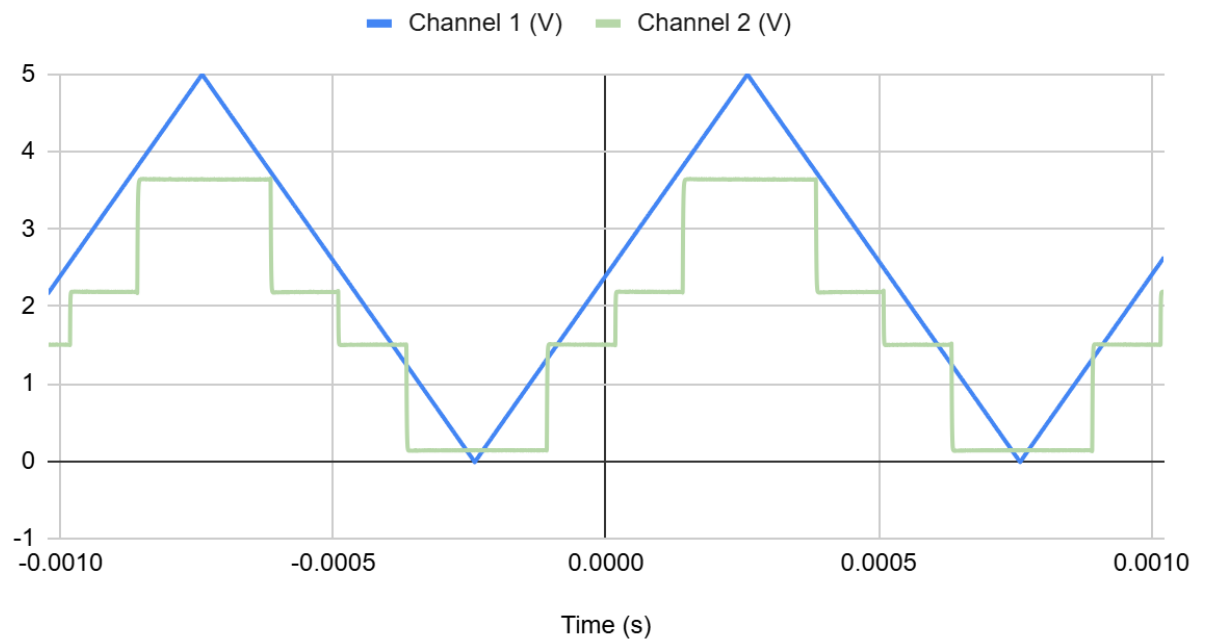


Figure 5.4: 2-bit DAC with Triangle Waveform at 1,000 hz

Part 6 – Integrator Filter

Integrator Filter

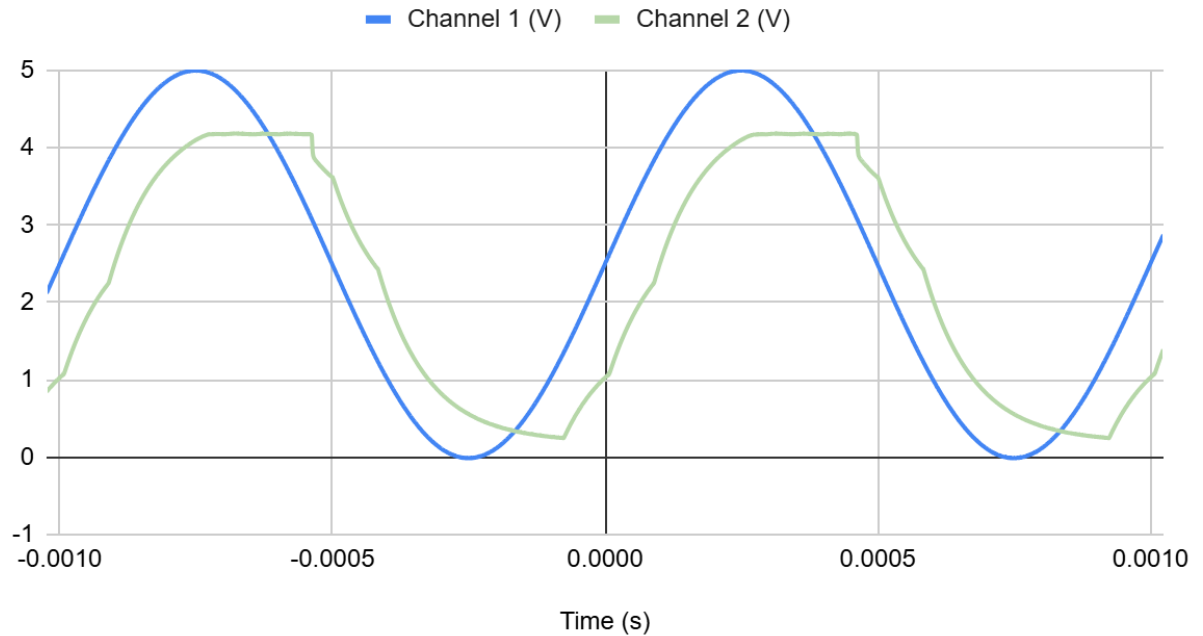


Figure 6.1: Integrator Filter

II. Discussion and Analysis

1-Bit ADC

At lower frequencies such as 500 Hz, the input triangle wave changes slowly enough for the comparator to reliably track the signal. This allows the 1-bit ADC to clearly distinguish between the high and low states without timing errors. The output transitions occur exactly when the input triangle wave crosses the reference voltage, producing a clean square wave that toggles between 0 V and 5 V, confirming the expected binary operation.

At low frequencies, the digital output is well aligned with the voltage-threshold crossings. However, as the frequency increases (1 kHz $\rightarrow$ 10 kHz), the comparator has less time to respond. The input crosses the reference voltage more rapidly, and the comparator's switching becomes delayed. This results in visible timing skew, slower transitions, and noticeable distortion in the digital output. Overall, higher frequency operation reveals the speed limitations of the 1-bit ADC and its comparator circuitry.

2-Bit ADC

The two digital output lines (DIOs) represent the quantized levels defined by the reference thresholds, creating the four valid binary states: 00, 01, 10, and 11. Each comparator changes to a HIGH state when the input voltage exceeds its corresponding threshold, generating a 2-bit code that maps the input signal to its quantized level.

At lower frequencies, the system exhibits accurate quantization with no missed transitions. The comparators have sufficient response time to sample the waveform and change states cleanly. As the frequency increases, the input signal crosses the thresholds more rapidly, and the comparators begin to struggle to synchronize. This produces conversion delays, occasional misalignment, and increased distortion in the quantized output streams. Higher frequency inputs expose the speed and resolution limitations inherent to the 2-bit flash-ADC structure.

R-2R DAC Output

Sine Wave Input:

For the 2-bit DAC, the four quantized levels (00, 01, 10, 11) generate a staircase approximation of the sinusoid. As the input sine wave rises, the DAC output steps upward through the quantized levels; as the sine wave falls, the DAC correspondingly steps downward. The staircase waveform closely follows the general shape of the sinusoid but with visible discretization due to the limited resolution.

Triangle Wave Input:

When applying a triangle wave, the rising slope causes the DAC output to climb through the quantized levels in evenly spaced steps. The falling slope produces the opposite effect. At higher frequencies, the number of steps per period increases, but the DAC has less time to switch states. This leads to transition errors, lag, and increased distortion in the reconstructed waveform. As frequency increases, the limited resolution and switching speed make the RMS error noticeably larger.

Integrator Filter Output

Before filtering, the DAC output appears as a staircase approximation of the input signal. After adding the integrator filter, the output becomes much smoother and more sinusoidal. However, the filtered waveform displays a flattened peak or plateau at the top of the sine wave. This occurs because the integrator does not respond quickly enough to reach the true maximum amplitude.

Additionally, the filter introduces phase lag, causing the output waveform to shift behind the original input. The slopes of the filtered waveform are less steep than the DAC steps, and the filter reacts more slowly to changes. Overall, the integrator reduces quantization noise and staircase artifacts but introduces amplitude limitations, phase shift, and some distortion due to its finite response time.